
INSTRUCTIONS TO THE COMPETITOR

Module E – Static Website Design

Project: ORION-X — Interplanetary Mission: Pathway to New Worlds

Introduction

Humanity's next great leap into deep space begins with ORION-X — an international mission tasked with charting a course from Earth to the outer solar system. The mission will travel past the Moon, on to Mars, and onward to the icy moons Europa and Titan, testing long-duration navigation, gravitational-assist trajectories, and deep-space communication systems along the way.

The ORION-X program office has assigned you the task of designing and building the mission's public-facing website: a place where the public can follow the journey, where researchers, engineers, and educators can learn how to get involved, and where partners can discover how to support the mission.

General Instructions

In this module, you must develop a static website for a client using only HTML and CSS.

No server-side or client-side framework is allowed for Module E.

CSS preprocessors may be used.

All HTML and CSS code, generated or hand-written, must pass the W3C HTML (validator.w3.org) and CSS (jigsaw.w3.org/css-validator) validators.

Competitor Information

Module E will be assessed using Google Chrome and Firefox.

The axe browser extension installed in Google Chrome will be used to assess that the website is “accessibility supported” according to WCAG.

Website Requirements

The client is ORION-X, an international deep-space exploration agency preparing to launch an interplanetary research mission. They have assigned you the task of implementing a promotional and informational website for the mission. The site must explain the mission to the general public, attract scientific, engineering and educational partners, and showcase the mission's planned route and current (simulated) status.

They have provided you with content for the website in the media files directory. Not all the assets have to be used. You can also create additional assets and text as you see fit. It is also possible to add links to pages that do not yet exist (for example, a press or media login page).

The website must be responsive and support at least the following viewports:

- Mobile portrait: 320×480 px
- Tablet portrait: 768×1024 px
- Desktop: 1280×800 px

As the website will be publicly available, it is essential to the client that the website conforms to accessibility guidelines (WCAG, at least level AA) and implements SEO best practices.

The following pages must be implemented.

Mission Control Page (Home Page)

Introduce the mission on the home page with the following essentials:

- Embedded launch/visualization video of the spacecraft (provided)
 - Autoplay when page is opened, without user interaction
 - Video is looped
 - Controls are hidden
 - Takes the full browser width while always maintaining the original aspect ratio
- Brief mission overview
- “Key Mission Facts” highlight strip
- Map / diagram showing the planned interplanetary route (Earth → Moon → Mars → Europa → Titan)
- Partner / sponsor logos

The Mission Control page must be the index page (index.html).

Suggested content / copy for this page

- **Hero headline:** “ORION-X: Pathway Beyond Earth”
- **Hero subtitle:** “An interplanetary mission charting a course through deep space — from the Moon to the icy moons of the outer solar system.”
- **Mission overview (suggested copy):** ORION-X is a long-duration spacecraft mission designed to test gravitational-assist trajectories, deep-space life-support systems, and interplanetary communication relays on a multi-year journey from Earth to Titan, with flybys of the Moon, Mars, and Europa along the way.

Key Mission Facts strip — suggested four short facts:

- Launch window: Q1 2027
- Mission duration: approx. 8 years
- Distance travelled: over 1.2 billion km
- Destinations: Moon · Mars · Europa · Titan
- **Route map caption:** “The planned trajectory of the ORION-X spacecraft across five major waypoints.”

ORION-
X

Mission
control

Interplanetary
route

Join
mission

Space data
center

autoplay

loop

no
controls



ORION-X: Pathway Beyond

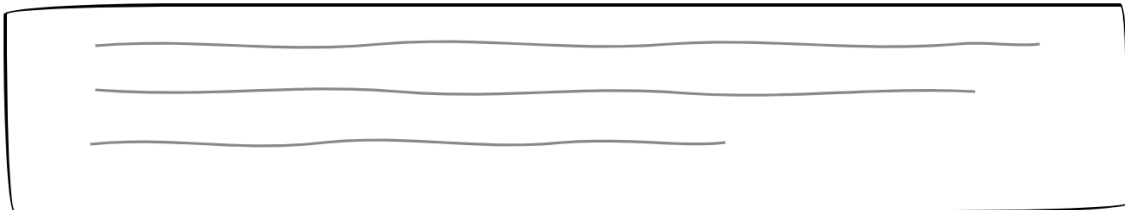
Earth

interplanetary exploration and simulation of deep-space travel
corridors

full browser width · aspect ratio
preserved

video: autoplay + loop + hidden controls via <video> attributes only — no JS needed

mission overview



key mission facts



launch 2027



~8 year duration

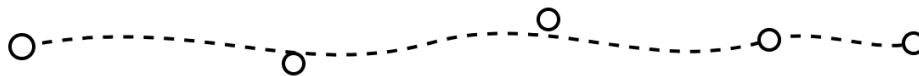


1.2B km total



4 waypoints

route map



earth → moon → mars → europa → titan

our partners

logo

logo

logo

logo

logo

Interplanetary Route Page

Once the spacecraft's full trajectory has been planned, visitors should be able to explore each leg of the journey and the destinations along the way.

The page must contain:

- A timeline of mission phases (Earth → Moon → Mars → Europa → Titan)
- Individual cards for each waypoint, each showing: distance from the previous waypoint, estimated travel time, and primary research objective
- An interactive way to select or highlight a waypoint card, implemented using HTML/CSS only (no JavaScript)
- A map or diagram showing the overall route (may reuse or extend the one shown on the Mission Control page)

Suggested content / copy for this page

- **Page heading:** “Interplanetary Route”
- **Intro paragraph:** *ORION-X follows a carefully staged trajectory, using the gravity of each waypoint to gain speed and conserve fuel on its way to the outer solar system.*

Suggested waypoint data (illustrative — may be adjusted, but should stay internally consistent across pages):

- Moon — ~384,000 km · ~3 days · gravity-assist maneuver and system checkout
- Mars — ~225 million km (avg) · ~7 months · atmospheric entry studies and sample-return relay testing
- Europa — ~628 million km · ~3 years cumulative · subsurface ocean and ice-shell investigation
- Titan — ~1.2 billion km · ~7 years cumulative · atmospheric chemistry and prebiotic molecule research

ORION-
X

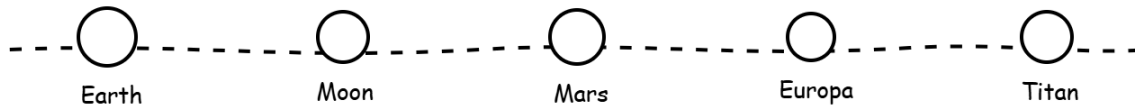
Mission
control

Interplanetary
route

Join
mission

Space data
center

Interplanetary Route



Moon

Distance: ~384,000 km

Travel time: ~3 days

Objective: gravity-assist + system checkout

Mars

Distance: ~225M km (avg)

Travel time: ~7 months

Objective: entry studies + sample relay

Europa

Distance: ~628M km

Travel time: ~3 yrs cumul.

Objective: subsurface ocean + ice shell

Titan

Distance: ~1.2B km

Travel time: ~7 yrs cumul.

Objective: atmosphere + prebiotic chem.



tapping / focusing a card expands it — done with `:checked`
or `:target`, no JS needed

Join Mission Page

Both organizations and private individuals should be able to become partners in the ORION-X mission.

It is possible to join the mission through multiple options:

- Sponsor a research instrument or onboard system
 - This allows the partner to provide a short text of at most 25 characters, or an organization logo, that will be featured on the mission's digital plaque
- Become a presenting partner (Ambassador)
 - This allows the partner to provide a logo that will be present on the website and on all official mission communication
- Register as an individual supporter (Explorer) and join the mission's digital archive

These different options have to be displayed on the page.

To be able to register interest in joining the mission, the page should contain a form to gather contact data (name, email, address, phone number) and it should allow the visitor to specify which participation role they choose (Explorer, Scientist, Engineer, or Ambassador).

Based on the chosen role, the specific fields for that option are presented.

Suitable form field types and validation must be chosen.

Handling of the form submission does not have to be implemented at the moment.

Suggested content / copy for this page

- **Page heading:** *“Join the Mission”*
- **Intro paragraph:** *ORION-X invites research institutions, companies, and individuals to take part in humanity's next great voyage.*

Role descriptions to display on the page:

- Explorer (individual supporter): “Register your name for inclusion in the mission's digital time capsule and follow ORION-X's journey.”
- Scientist / Engineer (instrument sponsor): “Sponsor a research instrument or onboard system — your name or logo will be featured on the digital mission plaque.”
- Ambassador (presenting partner): “Become the mission's presenting partner, with your logo featured across the website and all official ORION-X communications.”

Suggested form fields: full name / organization name, email address, postal address, phone number, role selection (radio buttons), and conditional fields — a text input with maxlength of 25 characters or a logo upload for the instrument-sponsorship option, a logo upload for the presenting-partner option, and a numeric/range input for a variable support amount for the individual-supporter option.

ORION-
X

Mission
control

Interplanetary
route

Join
mission

Space data
center

Join Mission

register as a partner and choose how you take part in ORION-X

choose your role

☐
Explorer☒
Scientist☐
Engineer☐
Ambassador

areas of collaboration

☒ engineering ☒ science ☐ education

partner registration

full name *

email address *

organization

phone number

 required + type="email" — the browser validates these fields automatically, no JS needed

Space Data Center Page

Once the mission launches, the public should be able to follow its progress. For this exercise, the page only needs to display static, illustrative values — no live data or scripting is required.

This page must contain a mission telemetry dashboard with at least the following data points:

- Mission status (e.g. “En Route to Mars”)
- Current objective (e.g. “Trajectory correction maneuver”)

-
- Distance to the next waypoint
 - Spacecraft power / energy level indicator
 - Communication delay / signal status

This page should also contain a map or diagram showing the spacecraft's current (simulated) position along the planned route, relative to the waypoints introduced on the Interplanetary Route page.

Suggested content / copy for this page

- ***Page heading:*** “Space Data Center”
- ***Intro paragraph:*** *This dashboard reflects ORION-X's current simulated status, updated throughout the mission for demonstration purposes.*

Suggested dashboard tiles:

- Mission status: “En route — Mars transfer orbit”
- Current objective: “Mid-course trajectory correction”
- Distance to next waypoint: “42.6 million km to Mars”
- Power level: “87% — nominal”
- Communication delay: “6 min 12 sec (one-way)”

ORION-
X

Mission
control

Interplanetary
route

Join
mission

Space data
center

Space Data Center

current (simulated) status of the orion-x spacecraft



mission status

en route — mars transfer
orbit



current objective

mid-course trajectory
correction



distance to object

42.6 million km to
mars



energy level

87% — nominal



communication status

6 min 12 sec delay (one-
way)



values shown are static placeholders for this demo — no live data or
scripting required

Global Elements

The following elements must be visible on all pages:

Navigation

- Must contain links to all four pages: Mission Control, Interplanetary Route, Join Mission, and Space Data Center
- Design must clearly highlight the current page
- Links must have adequate hover effects

Footer

- Must include at least a copyright notice
- Suggested additional content: mission name and tagline, quick navigation links, contact email, legal/disclaimer note

All elements can be enriched with more information where it makes sense.

Marking Summary

The submitted website will be assessed against the following sub-criteria. Total available marks: 20.

#	Sub-Criteria	Marks
1	Common elements (navigation & footer)	2.0
2	Mission Control page	4.0
3	Interplanetary Route page	3.5
4	Join Mission page	5.0
5	Space Data Center page	3.0
6	Standards (accessibility, validation, SEO)	2.5
	Total	20.0